Midterm Exam 2 (Version 2)

CAS CS 132: Geometric Algorithms

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- ▷ You will have approximately 75 minutes to complete this exam. Make sure to read every question, some are easier than others.
- ▷ Do not remove any pages from the exam.
- ▷ Your final solution must appear in the solution boxes for each problem. **Only include your final solution in the solution box. You must show your work outside of the solution box.** You will not receive credit if you don't show your work.
- ▷ We will not look at any work on the pages marked "*This page is intentionally left blank.*" You should use these pages for scratch work.

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1 True/False

(10 points) Determine if each of the following statements is **True** or **False**. Bubble in your answers below.

A. (1 point) For matrices A and B in $\mathbb{R}^{m\times n}$, let $\begin{bmatrix} A \\ B \end{bmatrix}$ denote the $\mathbb{R}^{2m\times n}$ obtained by stacking A and B vertically. For any vector $\mathbf{b} \in \mathbb{R}^n$, if $A\mathbf{x} = \mathbf{b}$ is consistent and $B\mathbf{x} = \mathbf{b}$ is consistent then so is $\begin{bmatrix} A \\ B \end{bmatrix} \mathbf{x} = \begin{bmatrix} \mathbf{b} \\ \mathbf{b} \end{bmatrix}$.

○ True

● False

B. (1 point) For any stochastic matrix $A \in \mathbb{R}^{n\times n}$, if A is regular then $\dim(\mathrm{Nul}(A - I)) = 1$.

● True

○ False

C. (1 point) For a subspace H of $\mathbb{R}^n$ and vector $\mathbf{v} \in \mathbb{R}^n$, if $\mathbf{v} \notin H$, then $\dim H < n$.

● True

○ False

D. (1 point) If A is invertible, then both the L-part and the U-part of an LU-factorization of A are invertible.

● True

○ False

E. (1 point) Given linear transformations $S : \mathbb{R}^m \to \mathbb{R}^n$ and $T : \mathbb{R}^n \to \mathbb{R}^m$, if $m < n$, then it is possible for $T \circ S$ to be invertible (recall that $(T \circ S)(\mathbf{v}) = T(S(\mathbf{v}))$).

● True

○ False

F. (1 point) If $\{\mathbf{v}_1, \mathbf{v}_2\}$ and $\{\mathbf{v}_1, \mathbf{v}_3\}$ and $\{\mathbf{v}_2, \mathbf{v}_3\}$ are linearly independent, then so is $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

○ True

● False

G. (1 point) For any stochastic matrix $A \in \mathbb{R}^{n\times n}$, if A has a unique steady-state distribution, then A is regular.

○ True

● False

H. (1 point) If $\operatorname{rank} A = 1$, then the matrix equation $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions.

○ True

● False

I. (1 point) For a matrix $A \in \mathbb{R}^{2026\times 2027}$, it is possible for the rank of A to be the same as to $\dim(\mathrm{Nul}\, A)$.

○ True

● False

J. (1 point) For matrices A, B, C in $\mathbb{R}^{n\times n}$, if $AB = AC$ then $B = C$.

○ True

● False

2 Matrix Inverses

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 1 & 0 & 2 \\ -2 & 5 & -7 \end{bmatrix}$$

A. (7 points) Determine the inverse of A, i.e., determine the matrix B such that $AB = BA = I$.

$$\left[\begin{array}{cccccc} 1 & -2 & 3 & 1 & 0 & 0 \\ \overset{-1}{1} & \overset{+2}{0} & \overset{-3}{2} & \overset{-1}{0} & 1 & 0 \\ \overset{+2}{-2} & \overset{-4}{5} & \overset{+6}{-7} & \overset{+2}{0} & 0 & 1 \end{array}\right] \sim \left[\begin{array}{cccccc} 1 & -2 & 3 & 1 & 0 & 0 \\ 0 & 2 & -1 & -1 & 1 & 0 \\ 0 & 1 & -1 & 2 & 0 & 1 \end{array}\right] \sim \left[\begin{array}{cccccc} 1 & -2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -1 & 2 & 0 & 1 \\ 0 & \overset{-2}{2} & \overset{+2}{-1} & \overset{-4}{-1} & 1 & \overset{-2}{0} \end{array}\right]$$

$$\sim \left[\begin{array}{cccccc} 1 & -2 & \overset{-3}{3} & \overset{+15}{1} & \overset{-3}{0} & \overset{+6}{0} \\ 0 & 1 & \overset{+1}{-1} & \overset{-5}{2} & \overset{+1}{0} & \overset{-2}{1} \\ 0 & 0 & 1 & -5 & 1 & -2 \end{array}\right] \sim \left[\begin{array}{cccccc} 1 & \overset{+2}{-2} & 0 & \overset{-6}{16} & \overset{+2}{-3} & \overset{-2}{6} \\ 0 & 1 & 0 & -3 & 1 & -1 \\ 0 & 0 & 1 & 5 & 1 & -2 \end{array}\right] \sim \left[\begin{array}{cccccc} 1 & 0 & 0 & 10 & -1 & 4 \\ 0 & 1 & 0 & -3 & 1 & -1 \\ 0 & 0 & 1 & 5 & 1 & -2 \end{array}\right]$$

Solution.

$$\begin{bmatrix} 10 & -1 & 4 \\ -3 & 1 & -1 \\ -5 & 1 & -2 \end{bmatrix}$$

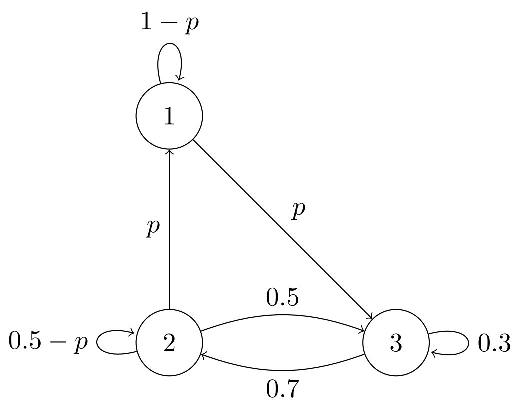
B. (3 points) Use the matrix you determined in Part A to calculate the solution to the matrix equation $A\mathbf{x} = [1\ 1\ 1]^T$. For partial credit, determine the solution to the matrix equation $A\mathbf{x} = [1\ 1\ 1]^T$ using Gaussian elimination.

$$\vec{x} = A^{-1}\begin{bmatrix}1\\1\\1\end{bmatrix} = \begin{bmatrix}10 & -1 & 4\\-3 & 1 & -1\\-5 & 1 & -2\end{bmatrix}\begin{bmatrix}1\\1\\1\end{bmatrix} = \begin{bmatrix}10 & -1 & +4\\-3 & +1 & -1\\-5 & +1 & -2\end{bmatrix} = \begin{bmatrix}13\\-3\\-6\end{bmatrix}$$

Solution.

$$\begin{bmatrix}13\\-3\\-6\end{bmatrix}$$

3 Markov Chains



A. (2 points) Determine the transition matrix A of the above state diagram in terms of p. The transitions out of state 1 should appear in the first column of A, out of 2 in the second column, and out of 3 in the third column.

Solution.

$$\begin{bmatrix} 1-p & p & 0 \\ 0 & 0.5-p & 0.7 \\ p & 0.5 & 0.3 \end{bmatrix}$$

B. (2 points) Is the matrix A regular in the case that $p = 0$? Justify your answer.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.5 & 0.7 \\ 0 & 0.5 & 0.3 \end{bmatrix}$$

Solution. ○ Yes ● No

Justification. $A\vec{e}_1 = \vec{e}_1 \Rightarrow A^K\vec{e}_1 = \vec{e}_1$ first col of A^k has zeros

C. (4 points) Determine $\lim_{k\to\infty} A^k[0.2\ 0.4\ 0.4]^T$ in the case that $p = 0.1$.

$$A = \begin{bmatrix} 0.9 & 0.1 & 0 \\ 0 & 0.4 & 0.7 \\ 0.1 & 0.5 & 0.3 \end{bmatrix} \qquad A - I = \begin{bmatrix} -0.1 & 0.1 & 0 \\ 0 & -0.6 & 0.7 \\ 0.1 & 0.5 & -0.7 \end{bmatrix} \sim$$

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 6 & -7 \\ \overset{-1}{1} & \overset{+1}{5} & -7 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 6 & -7 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{array}{c} \begin{array}{ccc} & +1 & -7/6 \end{array} \\ \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -7/6 \\ 0 & 0 & 0 \end{bmatrix} \end{array} \sim \begin{bmatrix} 1 & 0 & -7/6 \\ 0 & 1 & -7/6 \\ 0 & 0 & 0 \end{bmatrix}$$

$x_1 = 7/6\, x_3$

$x_2 = 7/6\, x_3$

x_3 is free

$$x_1 + x_2 + x_3 = \frac{7}{6}x_3 + \frac{7}{6}x_3 + \frac{6}{6}x_3 = \frac{20}{6}x_3 = 1$$

$x_1 = 7/20$

$x_2 = 7/20$

$x_3 = 6/20$

Solution.

$$\begin{bmatrix} 7/20 \\ 7/20 \\ 6/20 \end{bmatrix}$$

D. (2 points) Determine $\lim_{k\to\infty} A^k[0.2\ 0.4\ 0.4]^T$ in the case that $p = 0$. For partial credit, determine a general form solution that describes the steady-state vectors of A in the case that $p = 0$ (the steady state vectors of a matrix A are the vectors $\mathbf{v}$ such that $A\mathbf{v} = \mathbf{v}$).

$$A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.5 & 0.7 \\ 0 & 0.5 & 0.3 \end{bmatrix} \qquad A - I \sim \begin{bmatrix} 0 & 0 & 0 \\ 0 & -5 & 7 \\ 0 & 5 & -7 \end{bmatrix} \sim \begin{bmatrix} 0 & 1 & -7/5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

x_1 is free

$x_2 = 7/5\, x_3$

x_3 is free

$$\frac{7}{5}x_3 + \frac{5}{5}x_3 = 1$$

$$\frac{12}{5}x_3 = 1$$

$$x_3 = \frac{5}{\cancel{12}_{3}} \cdot \frac{\cancel{4}}{5} = 5/15$$

$$x_2 = \frac{7}{\cancel{12}_{3}} \cdot \frac{\cancel{4}}{5} = 7/15$$

$$x_1 = 1/5 = 3/15$$

Solution.

$$\begin{bmatrix} 3/15 \\ 7/15 \\ 5/15 \end{bmatrix}$$

4 Subspaces

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} \mapsto \begin{bmatrix} x + y - 3z \\ -x + 4z \\ 2x + y - 7z \end{bmatrix}$$

A. (2 points) Determine the matrix A that implements the above transformation, i.e., determine the matrix A such that the above transformation is equivalent to $\mathbf{x} \mapsto A\mathbf{x}$.

Solution.

$$\begin{bmatrix} 1 & 1 & -3 \\ -1 & 0 & 4 \\ 2 & 1 & -7 \end{bmatrix}$$

B. (4 points) Determine a basis for Nul A.

$$\begin{bmatrix} 1 & 1 & -3 \\ -1^{+1} & 0^{+1} & 4^{-3} \\ 2^{-2} & 1^{-2} & -7^{+6} \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & -3 \\ 0 & 1 & 1 \\ 0 & -1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1^{-1} & -3^{-1} \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$x_1 = 4x_3$
$x_2 = -x_3$
x_3 is free

$$x_3 \begin{bmatrix} 4 \\ -1 \\ 1 \end{bmatrix}$$

Solution.

$$\text{Nul}(A) = \text{span}\left\{ \begin{bmatrix} 4 \\ -1 \\ 1 \end{bmatrix} \right\}$$

You may also use this page to show your work for both Part B and Part C if necessary.

C. (2 points) Determine a basis for $\operatorname{Col} A$.

Solution.

$$\operatorname{Col} A = \operatorname{span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

D. (2 points) Let T denote the transformation from Part A and let H be the subspace

$$\left\{ \mathbf{v} : \mathbf{v} \in \operatorname{ran} T \text{ and } \mathbf{e}_3^T \mathbf{v} = 0 \right\}$$

That is, H is **the subspace of vectors in the range of T whose third components are 0.** Note that H is a subspace of $\operatorname{ran} T$, which is a subspace of $\mathbb{R}^3$. Determine a basis for H.

$$\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} - 2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix} \mapsto \begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix}$$

Solution.

$$H = \operatorname{span}\left\{ \begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix} \right\}$$

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